6 The probability distribution for the discrete random variable X is shown below.

x	0	1	2	3
$P(X = x)$	$3p^2$	$0.5p^2 + 2p$	$1.5p$	$1.5p^2 + 0.5p$

(a) Determine the value of p. **[4]**

(b) Determine the modal value of X. **[2]**